**What ethical responsibilities do software developers have when building proprietary surveillance tools?**

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RC 2001: Introduction to Writing Across Disciplines

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## Introduction to Question

### *Proprietary Software*

As this paper discusses the question "What ethical responsibilities do software developers have when building proprietary surveillance tools?" has been discussed and answered in the computer science field among researchers and professionals, it is important to clearly state the meaning of terms included in the question and bounds of the question. There is lively discussion slightly beyond the stated bounds, but for brevity, this literature review only concerns this question within the limits stated here.

The term "proprietary", when used in reference to software, is widely used to mean that the code for this software is not made freely accessible; the opposite term is "open-source". If the code is open-source, several properties intuitively follow. Such properties include customizability, transparency, and no cost usage. Transparency and customizability are the most relevant properties for this review's consideration.

One can imagine how, when software is open-source, sufficiently knowledgeable observers can interpret the code and explain exactly how it works and what it does; these people can then share their conclusions. In this way, (nearly) everyone affected by the technology can understand it. In this way, proprietary software asserts a power dynamic in its secrecy.

Additionally, open-source software can be customized as freely as one is able to develop the desired extensions or changes to the code. While this allows small customizations, major changes can also be made which shift the application of the software or remove guardrails. One can see that proprietary software, then, has a more controlled lifecycle.

Because of these properties, the anticipations of a developer regarding their software's use and legacy are highly dependent on whether the software is proprietary. Developer responsibility is not necessarily less or more, but different. Proprietary developers have more control over software usage and transparency. It should be noted though that when stating that a developer can keep control in these ways, there is an unstated caveat that they would need some level of power in their respective organization (consider the intern vs the CEO)  Extensive discourse regarding open-source developer responsibilities, including in surveillance, exists, but this is not included in this review.

### *Intended Use*

This review only concerns discussion of cases where the developer intends their software to be used for surveillance. Proprietary software can be used in ways the developer does not intend. For example, Kaster S. & Ensign P. of Lazaridis School of Business and Economics (2022) in their case study, "Privatized espionage: NSO Group Technologies and its Pegasus spyware", examine how a software designed for counterterrorism and criminal investigation was used by governments to surveil journalists and dissidents. Consideration of cases like this implies added complexity to answering the titular question when considering unintended usage of software. For brevity, the reviewed discussion here only concerns intended usage of proprietary software.

### *Scope of Sources*

Additionally, this review largely focuses on discussion within the computer science community, so legal regulations will not be directly reviewed.

## Foundational Ideas

***Technology and Politics***

Professor of Social Science and Technology, Winner L. (1980), in widely cited article "Do Artifacts Have Politics", addresses the issue of technology having inherent "political qualities", arguing the affirmative. Winner explains two possible ways an artifact can be political: through choices in its flexible design, and inherently suggestive of a political system.

For the former, Winner gives the example of bridges designed such that they were too low for public buses to pass. In this case, he claims this design was of malign, but he also discusses designs injected with politics by systemic means; specifically, he claims that in university's development of agricultural technology, developers implicitly imbue bias toward large farming schemes.

For the latter type of political artifact, he cites an example discussed by both Plato and Engels: a ship. The design of a ship, one that requires steady control, is suggestive of a political system aboard – one captain and an obedient crew.

This work is highly relevant to the discussion of relationship between developer, technology, and usages of that technology. The titular discussion lies in part on that broader discussion's foundation, narrowing the focus to surveillance technology.

***Surveillance***

Computer scientist and privacy advocate, Clarke R., (1987) in cornerstone article "Information Technology and Dataveillance", introduces the concept of "dataveillance". Clarke defines this term as "systematic monitoring of people's actions or communications through the application of information technology". He distinguishes dataveillance from physical or communication surveillance (e.g. directly watching and telephone taps, respectively) as superior in efficiency.

Clarke states that IT professionals and academics have a responsibility to "appreciate the power" of created or studied technology. Clarke writes that professionals and academics should advocate and develop safeguards as well as inform relevant parties of harmful effect which their technology causes. Clarke expresses a weaker stance on developer responsibility, though, than Winner (1980). He writes, "It would be inappropriate for the purveyors of any technology to be responsible for decisions regarding its application." but also dismisses inaction.

## Landscape and Approaches

### *Internal Ethics*

Data researchers, Metcalf, Moss, & Boyd (2019) in article "Owning Ethics: Corporate Logics, Silicon Valley, and the Institutionalization of Ethics", examine internal ethics in big tech companies. They do this via interviews with 17 "ethics owners" at large tech companies. "Ethics owner" is a job position responsible for managing ethics within the organization. The authors describe a tension between their roles of bringing external perspectives to the organization and being intimately involved with operations.

The authors identify three main principles that motivate Silicon Valley ethics: meritocracy (that technically skilled people are best suited to deal with ethics), technological solutionism (that technology can be used to solve ethics problems), and market fundamentalism (that market forces govern possibilities foremost). The authors claim these principles often serve to affirm practices, rarely challenging them.

This article gives insight into how internal ethics regarding issues like surveillance are managed at big tech companies. According to the authors of this article, such internal ethics are ineffectual and largely for performance.

### *Contemporary Surveillance Offenses*

Privacy researcher and advocate Deibert R. (2022) in article “Subversion Inc: The Age of Private Espionage”, discusses the landscape of private surveillance technology. He describes how private companies develop technology which they sell to human rights abusing government and private entities, among other clients. Deibert also characterizes the legal nature of this as near impunity because of national security exceptions, nondisclosure agreements, and complex ownership structures.

Deibert recommends stronger legal bounds on the private surveillance industry including prosecution, export policies, and transparency of government surveillance practices and adoptions. Deibert argues for the importance of these measures in preserving rule of law and liberal democracy.

## Policies

### *ACM 1999*

Executive committee members of IEEE-CS/ACM Joint Task Force on Software Engineering Ethics and Professional Practices Gotterbarn, Miller, & Rogerson (1999) in article “Computer Society and ACM Approve Software Engineering Code of Ethics”, review Version 5.2 of Software Engineering Code of Ethics and Professional Practice.

Important changes from the last version include indicated priority for public over product, significant for direction of developer responsibility. The new ethics also make it clear that the public's interests are above that of employers or clients.

Additionally, the ethics command whistleblowing where appropriate. This is phrased, "Disclose to appropriate persons or authorities any actual or potential danger to the user, the public, or the environment" (p. 86, Principle 1.04). This inclusion is highly relevant to surveillance technology.

Other important parts refer to managers' obligation to not force employees to violate ethics, identifying and documenting social concerns, and the inclusion of privacy in product design. This code of ethics is highly significant to the timeline of surveillance ethics.

***ACM 2018***

ACM Code 2018 Task Force (2018) in the current "ACM Code of Ethics and Professional Conduct", specifies the current ethical code mandated by ACM. Developer responsibilities, leadership responsibilities, and general principles are major focuses.

The code maintains the 1999 emphasis that the public is the top-most priority for developers. Also, this code of ethics addresses the modern state of technology, recognizing that many key decisions like hirings and firings are made using technology. The whistleblower obligation is also maintained.

The code of ethics commands that as little personal information be collected as possible. Also, this data may only be used for purposes for which users give their consent. Developers are also urged to consider how data aggregation may be used with other technologies.

Overall, the 2018 code of ethics carries a stronger declaration of developer responsibility than the 1999 code, though many principles remain the same. This is likely due to the greater role computers and other technologies have in societal structure.

***AINow 2025 Landscape Report***

Brennan, Kak, and West (2025) in the highly comprehensive report “Artificial Power”, provide a highly political review of the AI landscape, with several references to surveillance technology.

The authors write that AI is an “inherently centralizing technology” (p. 14) that is constantly used for surveillance and military, suggestive of Winner’s second type of political artifact, that which is highly suggestive to a political structure.

Additionally, the authors condemn use of AI for worker surveillance to increase productivity, citing use by Amazon in warehouses. They also note Google’s February 2025 reversal in their guidelines, allowing their AI to be used for surveillance and military, and opposing responses to this within and without the company.

The authors adamantly assert that tech companies cannot self-regulate AI, pointing to the low incentive they must uphold promises, allowing for cases like Google’s guidelines reversal. The authors assert the need for government regulation.

Among prescribed regulations are strict limits on the use of surveillance to adjust prices and wages. Also, the authors write that it should be illegal to sell worker information from surveillance to third parties.

This report is much more political than the ACM codes, though, notedly, the 2018 report is more strongly worded than the 1999 version. This is explainable by the publishing organization. AI Now is a civil research institute, and the ACM/IEEE are professional organizations. Still, the timeline created by these three documents may indicate intensifying discussion about ethics in technology including surveillance ethics.

## Conclusion

The question of focus in this paper has been discussed since the 1980s and has evolves with the course of relevant technological developments. Whether the technology or ethics discussion led this co-evolution is unclear, but some experts like Deibert and the authors of the AI Now report indicate that in recent times at least, though maybe also in decades past, ethics and regulations have not kept up. Others like Metcalf et al. indicate insufficiency of current ethics, particularly internal ethics. Also, these sources may indicate a growing intensity in the discussion.

There are multiple fronts in addressing this issue. There is the front of internal ethics, that of professional codes of ethics, and that of government regulation. Past and present government regulations have not been addressed in this essay for purposes of scope discussed in the introduction. It seems the near future of the discussion is dependent on the individual, and perhaps, coordinated development of these fronts.

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